

Phase-change material-based external cavity lasers for optical communications

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Abstract: We propose an external-cavity laser architecture that employs an external silicon nitride (Si_3N_4) waveguide sidewall Bragg grating integrated with a phase-change material, specifically antimony trisulfide (Sb_2S_3). Leveraging the significant contrast in the effective refractive index between the amorphous and crystalline states of Sb_2S_3 , along with its non-volatile phase transition property, this design enables stable, selective lasing at target wavelengths across a broad frequency tuning range. Simulation results indicate that the wavelength tuning range of this structure can cover the entire C-band, with a channel spacing of ~ 0.8 nm between adjacent wavelengths. This external-cavity laser offers an efficient solution for optical communications, exhibiting strong potential for enhancing precise wavelength tuning capability, network flexibility, and resource utilization efficiency.

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1. Introduction

Wavelength Division Multiplexing (WDM) technology has revolutionized optical fiber communications by enabling the simultaneous transmission of multiple optical carrier signals on a single fiber [1–4]. This technology significantly increases the transmission capacity of communication systems, laying the foundation for multi-dimensional multiplexing technology and ultra-large-scale data center interconnection for future 6G evolution [3,5–8]. However, the light source is required to be able to freely switch wavelengths across the entire C-band in the WDM system. Against this backdrop, the development of versatile tunable lasers with a wide tuning range is essential to meet the need for generating all subcarriers, which can significantly reduce the cost of optical sources [9–11]. Moreover, tunable lasers are instrumental in diverse fields: dynamic routing and wavelength allocation in optical communication, high-resolution optical coherence tomography (OCT) for biomedical imaging, precise spectral analysis of pollutants in environmental monitoring, and frequency-modulated light sources for coherent lidar in autonomous driving.

Tunable lasers are commonly implemented through several principal architectures, such as distributed feedback (DFB) laser arrays, distributed Bragg reflector (DBR) lasers, and external-cavity lasers (ECLs) employing spatial diffraction gratings, and heterogeneous integrated ECLs. Among these, DBR lasers employ two mirrors with slightly different periodicities to achieve an extended tuning range via the vernier effect. While offering advantages such as rapid tuning speed and full electrical control, their performance is constrained by linewidth broadening due to free carrier absorption. Conversely, DFB laser arrays integrate multiple DFB lasers emitting at distinct fixed wavelengths in parallel, with their outputs combined via a multiplexer. This approach benefits from technological maturity and reliability but suffers from high power consumption and a large footprint. ECLs based on spatial light diffraction gratings utilize microelectromechanical systems (MEMS) or other mechanical actuators to alter the optical path angle for wavelength

selection. Although capable of a broad tuning spectrum, these systems are typically bulky and challenging to integrate monolithically. Alternatively, heterogeneously integrated ECLs offer a compelling solution by combining gain chips with high-performance passive feedback elements on silicon-based platforms, with their performance pivoting primarily on the properties of these feedback components, including microrings and Bragg gratings. In particular, ECLs employing waveguide Bragg gratings as the external cavity combine a wide frequency tuning range, narrow linewidth, and excellent side-mode suppression, offering a scalable technological route toward integrated and compact laser sources [12]. Waveguide sidewall Bragg gratings also have attracted considerable research attention because of their compact geometry, ease of integration, and compatibility with standard lithographic and etching processes, making them cost-effective and suitable for mass production [13].

Conventional tuning mechanisms for tunable photonic devices mainly depend on the effects of thermo-optic or electro-optic [14]. Thermo-optic tuning is limited by the thermal relaxation time of the material, resulting in relatively slow response speeds [15]. Electro-optic tuning, while offering faster response capabilities [16], typically requires a continuous bias voltage, leading to significant static power consumption [17]. These factors collectively restrict the potential of traditional tuning schemes for further application in high-speed, low-power photonic systems. Phase-change materials (PCMs), with their non-volatility, high refractive-index contrast, nanosecond-scale switching speed, and CMOS-process compatibility [18], offer a compelling alternative. They can overcome the slow response of thermo-optic modulation and eliminate static power consumption in the tuned state by leveraging their non-volatile nature. The concept of tuning Bragg gratings with PCMs has been validated. In 2022, Gao et al. integrated Sb_2Se_3 into Bragg gratings and demonstrated a hybrid Sb_2Se_3 -Si Bragg grating for band-stop or narrow-bandpass filtering. In a uniform grating, they achieved a 12 nm stop band with 17 dB extinction ratio and an 18 nm stop-band shift via phase-change control of Sb_2Se_3 [19]. It provides an innovative pathway for waveguide-Bragg-grating-based ECLs that simultaneously achieves fast tuning and zero standby power. Although conventional chalcogenides such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$ exhibit high propagation loss in the C-band, recently reported optical PCMs like Sb_2Se_3 and Sb_2S_3 not only offer low insertion loss in both two states but also exhibit a strong disparity in refractive index [20]. They enable sub-microsecond, low-energy phase transitions, making them highly suitable for the design and implementation of dynamic photonic devices.

This paper proposes a novel external-cavity laser architecture based on a silicon nitride waveguide sidewall Bragg grating featuring an integrated phase-change material Sb_2S_3 , enabling broadband frequency tuning via electrically induced Joule heating. The design fully exploits the large effective refractive index contrast between the amorphous and crystalline states of Sb_2S_3 , its fast reversible switching characteristics, and its non-volatile nature, allowing precise selection and stable output of the target wavelength over a broad frequency tuning range. Simulation results demonstrate that the structure achieves continuous wavelength tuning across the C-band (1530 nm-1564.4 nm) with a channel spacing of 0.8 nm. This external-cavity laser provides an efficient wavelength control solution for optical communication systems and is expected to enhance WDM systems by enabling flexible channel configuration and improving resource utilization.

2. Device structure and principle

Figure 1 illustrates the structure of the silicon nitride (Si_3N_4) sidewall-grating external-cavity laser integrated with Sb_2S_3 thin film. As shown in Fig. 1(a), the proposed ECL comprises three key sections: the semiconductor optical amplifier (SOA) acting as both the gain medium and seed source, the phase section, and the external optical reflector. The phase section is formed by isolating a 150 μm -long waveguide section on the ridge waveguide of the SOA. The external optical reflector is composed of a Si_3N_4 waveguide Bragg grating (WBG) integrated with an

Sb_2S_3 layer, enabling wavelength-selective reflection and electrically tunable resonance through refractive-index modulation. Figure 1(b) presents a cross section of the WBG-based reflector structure. Along the z -axis, the layers are arranged as follows: a Si_3N_4 WBG with the waveguide thickness $h_s = 330$ nm, the grating depth $h_G = 330$ nm, and the period Λ , an indium-tin-oxide (ITO) layer with the thickness $h_{\text{ITO}} = 20$ nm, and an Sb_2S_3 layer with the thickness h_{pcm} . Figure 1(c) shows the top-view layout of the ECL. A pair of palladium (Pd) electrodes deposited above the ITO layers on both sides of the waveguide provides electrical excitation. A needle-array contact scheme is adopted to mitigate the non-uniform temperature distribution caused by localized Joule heating. Specifically, amorphization of Sb_2S_3 is triggered by high-amplitude, short-duration electrical pulses, whereas crystallization is driven by low-amplitude, long-duration pulses. The ridge waveguide width on the SOA is set to $W_a = 2$ μm , the Si_3N_4 WBG width to $W_s = 1.3$ μm , and the sidewall grating width to 100 nm.

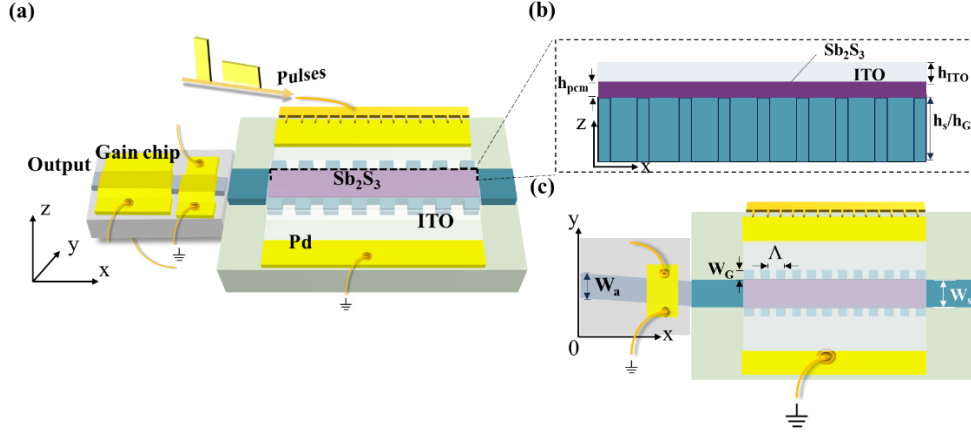


Fig. 1. (a) Schematic 3D illustration of the proposed ECL structure. (b) Cross-sectional view of the external-cavity waveguide. (c) Top-view layout of the ECL.

The operating principle of the proposed ECL is as follows. Firstly, a drive current is injected into the SOA, generating initial seed light. The light is then coupled into the WBG region integrated with the Sb_2S_3 phase-change material. In this region, a bias voltage is applied to the Pd electrodes to generate Joule heating, which is conducted through the transparent ITO layer to the Sb_2S_3 film. Secondly, the Joule heating triggers a reversible phase transition of Sb_2S_3 between its amorphous and crystalline states, accompanied by a pronounced change in its refractive index, thereby enabling coarse wavelength tuning of the laser. An electrode is also placed above the phase section. By electrically heating this region, the phase can be finely adjusted, which not only facilitates fine wavelength tuning but also effectively suppresses multimode oscillation of the laser [21]. Finally, the light reflected by the WBG is recoupled into the active region of the SOA. As a result, a laser output at the front facet with a narrow linewidth can be achieved.

3. Device design and simulation results

At the 1550 nm operating wavelength, the effective refractive indices of Sb_2S_3 in the amorphous and crystalline states are $n_{\text{a-Sb}_2\text{S}_3} = 2.85$ and $n_{\text{c-Sb}_2\text{S}_3} = 3.2$, respectively [22]. Notably, Sb_2S_3 exhibits nearly zero optical absorption ($\text{Im}(n) \approx 0$) in this spectral range, which significantly reduces waveguide propagation loss. To achieve broadband wavelength tuning of the laser, the difference in refractive-index of the PCM between its amorphous and crystalline states must be enlarged. To achieve this purpose, the PCM cladding layer's thickness was set to 55 nm. The finite-difference eigenmode (FDE) simulations were performed to analyze the electric-field

distribution in the hybrid waveguide under different phase states of Sb_2S_3 . As shown in Fig. 2, when Sb_2S_3 is in the crystalline state, the electric field shifts upward, confirming that the difference in refractive-index between the two states perturbs the optical confinement and thereby modulates the Bragg resonance.

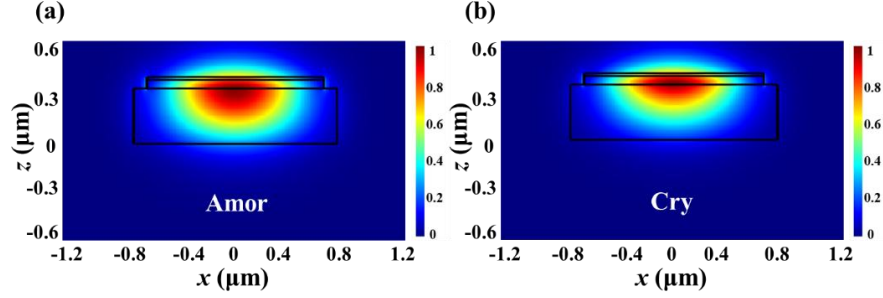


Fig. 2. Distribution of electric field in the hybrid waveguide with Sb_2S_3 in different phase states: (a) amorphous state, (b) crystalline state.

Theoretical analysis indicates that when the multiplication of the grating coupling coefficient (κ) and the grating length (L_G) exceeds a critical threshold (typically $\kappa L_G > 2$) [23], the likelihood of this phenomenon increases significantly. The coupling coefficient κ can be calculated using the following formula:

$$\kappa = \frac{\pi \Delta n}{\lambda_{\text{Bragg}}} \quad (1)$$

where, Δn represents the effective refractive index difference between the grating ridge and groove regions, and λ_{Bragg} denotes the Bragg wavelength.

We compared the relationship between the h_{pcm} , and several key optical parameters under different phase states. These parameters include: Δn (Fig. 3(a)), κ (Fig. 3(b)), the waveguide effective refractive index (n_{eff}), and the difference in between the amorphous and crystalline states of Sb_2S_3 (Δn_{eff}) (Fig. 3(c)). Here, Δn determines the grating coupling coefficient κ , n_{eff} determines the central reflection wavelength of the laser, and Δn_{eff} determines the tunable range of the laser. As shown in Fig. 3, Δn and κ exhibit a negative correlation with h_{pcm} , while n_{eff} and Δn_{eff} exhibit a positive correlation. This behavior is primarily attributed to the competition mechanism between optical field confinement and material refractive index contrast within the waveguide structure. When h_{pcm} is small, the optical field is mainly confined within the Si_3N_4 waveguide core layer, resulting in a large effective refractive index difference between the ridge and groove regions, and consequently, higher values of Δn and κ . As h_{pcm} increases, the PCM layer exerts a greater influence on the waveguide refractive index, and the optical field distribution shifts toward the PCM layer, diluting the refractive index contrast between the ridge and groove regions, which leads to a decrease in Δn and κ . Meanwhile, because the refractive index of the PCM material (especially in the crystalline state) is relatively high, increasing its thickness enhances the overall effective refractive index n_{eff} . At the same time, the refractive index contrast between the amorphous and crystalline states is more fully realized at larger thicknesses, so Δn_{eff} increases with h_{pcm} . These results indicate that a trade-off between contra-directional coupling intensity and tuning range must be balanced in device design.

When the deposited thickness of Sb_2S_3 is 55 nm, the effective refractive index change of the Sb_2S_3 -WBG is $\Delta n = 0.00943$ when Sb_2S_3 is in the amorphous state and $\Delta n = 0.00761$ when it is in the crystalline state. The corresponding calculated κ values are 191 cm^{-1} for the amorphous state and 154 cm^{-1} for the crystalline state. The transition between the amorphous and crystalline states induces an effective refractive index contrast of approximately 0.08. This level of contrast

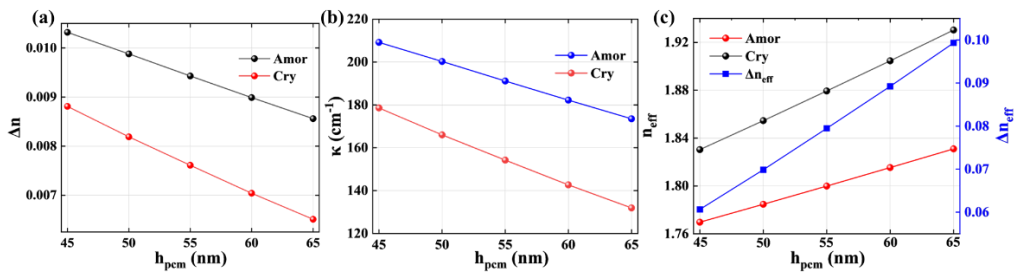


Fig. 3. Key optical parameters as a function of PCM deposition thickness (h_{pcm}). (a) Δn ; (b) κ ; (c) n_{eff} and Δn_{eff} , where Δn determines κ , n_{eff} determines the reflection wavelength, and Δn_{eff} determines the tuning range

is sufficient to support wavelength tuning across the entire C-band for the proposed ECL structure. Therefore, we selected a deposition thickness of 55 nm for Sb_2S_3 . When the material is in the amorphous state, the effective refractive index of the waveguide is 1.8. According to the grating period formula $\lambda = 2n_{\text{eff}}\Lambda$, the calculated grating period is 425 nm.

Single-longitudinal-mode output of the ECL is governed by the product of the coupling coefficient κ and the grating length L_G . With κ fixed by the preceding design, the judicious choice of L_G becomes pivotal for suppressing the side mode. In principle, L_G should be made as short as possible to shrink the κL_G product; yet the grating length also sets the threshold current of the laser, and an excessively short L_G will raise that threshold. Moreover, thermal effects further complicate the performance trade-off. We therefore systematically mapped the dependence of threshold current on grating length under different temperatures, seeking the optimum balance between thermal effect and low-threshold operation; Table 1 lists the simulation parameters for the proposed ECL which employed in the simulation.

Table 1. Simulation parameters for the proposed ECL

Symbol	Description	Value
L_a L_p	Length of active section Length of phase section	200 μm 150 μm
W	Active region width	2 μm
T_a	Active region thickness	60 nm
A	Linear recombination factor	$1 \times 10^8 \text{ s}^{-1}$
B	Bimolecular recombination factor	$1 \times 10^{16} \text{ m}^3 \text{ s}^{-1}$
C	Auger recombination factor	$3.5 \times 10^{-41} \text{ m}^3 \text{ s}^{-1}$
$n_{\text{eff}0}$	Effective refractive index at carrier transport concentration	3.2
N_g	Group refractive index	3.6
N_T	Transparent carrier concentration	$1 \times 10^{24} \text{ m}^{-3}$
σ	Linewidth broadening factor	1
Γ	Optical field confinement factor	0.079
gN	Differential gain	$1.5 \times 10^5 \text{ m}^{-1}$
T_{gain}, T_d	Characteristic temperatures	200 K
T_{re}	Reference temperature	300 K

In the dynamic model, according to the coupled-mode equations and the carrier-rate equations [24], the forward-wave electric field (F) and the backward-wave electric field (R) satisfy the

following equations:

$$\frac{1}{v_g} \frac{\partial F(z, t)}{\partial t} + \frac{\partial F(z, t)}{\partial z} = \left(\frac{\Gamma g - \alpha_s}{2} - i\delta \right) F(z, t) - i\kappa R(z, t) + S_F \quad (2)$$

$$\frac{1}{v_g} \frac{\partial R(z, t)}{\partial t} + \frac{\partial R(z, t)}{\partial z} = \left(\frac{\Gamma g - \alpha_s}{2} - i\delta \right) R(z, t) - i\kappa F(z, t) + S_R \quad (3)$$

where F and R denote the amplitudes of the forward-wave and backward-wave electric fields, respectively, representing the proportion of the optical field intensity in the active region that occupies the entire transverse cross-section of the laser. v_g is the group velocity, g is the material gain, α_s denotes the absorption-scattering loss, δ represents the detuning factor, Γ is the optical confinement factor, S_F and S_R are the spontaneous emission noise terms for the forward and backward waves, respectively, and κ represents the grating coupling coefficient.

Using the Split-step method [25], the laser cavity length can be discretized into many small segments of length Δz , each corresponding to a propagation length of $v_g \Delta t$ in the time domain. Consequently, the time axis can be divided into discrete steps Δt , allowing Eqs. (2) and (3) to be rewritten in the following discretized form:

$$F(z + \Delta z, t + \Delta t) = \exp \left(((\Gamma g - \alpha_s)/2 - i\delta) \Delta z - i \frac{\Delta \varphi}{2} \right) F(z, t) - i\kappa \Delta z R(z + \Delta z, t) + S_F \Delta z \quad (4)$$

$$R(z + \Delta z, t + \Delta t) = \exp \left(((\Gamma g - \alpha_s)/2 - i\delta) \Delta z - i \frac{\Delta \varphi}{2} \right) R(z + \Delta z, t) - i\kappa \Delta z F(z, t) + S_R \Delta z \quad (5)$$

where $\Delta \varphi$ denotes the phase shift caused by the grating. In practical simulations, we can evaluate the photon density, carrier concentration, material gain, effective refractive index, and other relevant parameters at a given time step. By solving Eqs. (4) and (5), the values of these parameters for the next time step are obtained. This iterative process is repeated continuously until steady-state lasing is achieved. Throughout the iteration, the output power of the laser at each time interval can be monitored.

Thermal effects alter the carrier density and temperature distribution in the active region of semiconductor lasers, thereby modulating the material gain coefficient (g) and the effective refractive index (n_{eff}) of the waveguide. This ultimately leads to variations in key laser performance parameters, including the output power and threshold current, as illustrated in Eqs. (6) and (7).

$$g = \frac{g_N \exp[-(T - T_{\text{re}})/T_{\text{gain}}] \{N - N_T \exp[(T - T_{\text{re}})/T_d]\}}{2(1 + \varepsilon P)} \quad (6)$$

$$n_{\text{eff}} = n_{\text{eff}}^0 - \Gamma \sigma g \frac{\lambda_r}{4\pi} + \alpha(T - T_{\text{re}})$$

where T represents the average temperature of the active region, and T_{re} is reference temperature, T_{gain} , T_d are the thermal characteristic temperatures for the differential gain and transparent carrier density, respectively. At the reference temperature T_{re} , the differential gain takes the value g_N , and the transparent carrier density takes the value N_T . In addition, the gain compression factor is represented by ε . n_{eff}^0 denotes the refractive index at T_{re} , λ_r is the reference wavelength, σ is the linewidth enhancement factor. The temperature dependence of the change of effective refractive index due to thermal processes is described by the coefficient α .

Figure 4(a) illustrates the relationship between the laser threshold current and the grating length under thermal effects at 300 K. As reported in other study, an inverted taper structure has been employed as the coupling interface between the SOA and the Si_3N_4 waveguide. Experimentally measured coupling losses in such configurations can reach approximately 3 dB [26]. Therefore, we have incorporated the coupling loss into the present model based on the current estimation

derived from experimentally validated literature values. The power-current (P-I) curve exhibits pronounced nonlinear saturation characteristics, with the threshold current being inversely proportional to the grating length, as shown in the inset of Fig. 4(a). When thermal effects at 350 K are considered (Fig. 4(b)), the nonlinear saturation behavior of the P-I curve becomes even more pronounced. This saturation phenomenon primarily arises from the temperature-induced reduction in the gain coefficient, which leads to output power saturation [27]. Notably, the inset results in Fig. 4(a) and (b) indicate that under the present thermal conditions, devices with grating lengths of 500 μm and 700 μm do not exhibit significant differences in threshold current. Considering that a lower threshold current is of great significance for reducing the overall power consumption of the laser [26], and that a shorter grating length is conducive to improving device integration, a grating length of 500 μm was ultimately selected for the waveguide Bragg grating in this study. On the basis of Finite-Difference Time-Domain (FDTD) simulation calculations, the corresponding grating reflectivity is 72%.

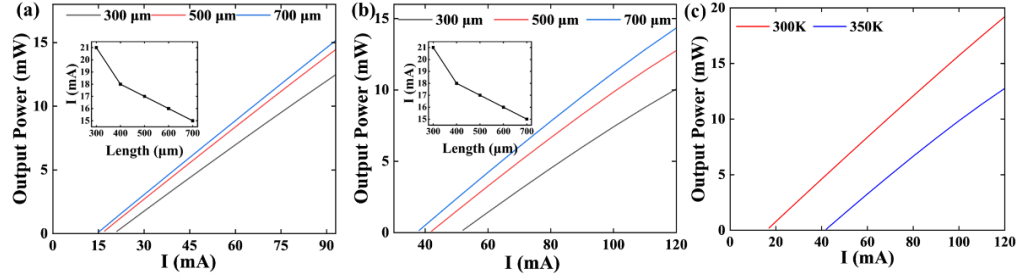


Fig. 4. Relationship between grating length and P-I curves under different thermal conditions: (a) with thermal effects at 300 K, (b) with thermal effects at 350 K. The insets show the corresponding relationship between grating length and laser threshold current for each thermal condition. (c) Power-current (P-I) characteristic curve of the laser at thermal conditions of 300 K and 350 K with a grating length of 500 μm .

We further calculated the P-I characteristics of the laser at 300 K and 350 K for a grating length of 500 μm , with the results presented in Fig. 4(c). The analysis confirms that thermal effects have a considerable impact on the laser output power: stronger thermal effects lead to a higher threshold current and lower output power. It is important to emphasize that the phase transition of the Sb_2S_3 film deposited on the external cavity surface is intrinsically a thermally driven process. The transition from the crystalline to the amorphous state is achieved by applying a high-amplitude, short-duration voltage pulse to the Pd electrodes. The resulting Joule heating rapidly raises the temperature of Sb_2S_3 above its melting point ($T_m \approx 801$ K) [28], followed by rapid quenching to form a disordered amorphous structure. In contrast, the transition from the amorphous to the crystalline state needs a low-amplitude, long-duration voltage pulse, which raises the temperature of the Sb_2S_3 to between its glass-transition temperature ($T_g \approx 573$ K) [28] and its melting point, thereby enabling atomic rearrangement into an ordered crystalline lattice. Given that both phase-change processes involve substantial heat generation, it is crucial to prevent thermal crosstalk to the semiconductor SOA during PCM heating to minimize thermal interference with laser performance and reduce the overall impact of thermal coupling on the device.

Furthermore, the laser linewidth is also an important metric for evaluating laser performance, Henry *et al.* have derived a widely used expression for the linewidth broadening of external-cavity semiconductor lasers [29]:

$$\Delta\nu = \frac{\Gamma v_g^2 h\nu_g n_{sp} \alpha_m (1 + \sigma^2)}{8\pi P_0} \frac{1}{[1 + \tau_{ex}/\tau_{in}(1 - r_1/r_2)]^2} \quad (8)$$

where Γ is the optical confinement factor, v_g is the group velocity, h is Planck's constant, ν is the frequency of the laser, n_{sp} is the spontaneous emission factor, α_m is the facet loss coefficient, σ is the linewidth enhancement factor, and P_0 is the output power. τ_{ex} and τ_{in} are the round-trip times in the external cavity and gain region, respectively, while r_1 and r_2 denote the reflectivity of the external reflector and the back facet of the gain section.

The grating delay characteristic τ_{ex} can be obtained through the transfer matrix, and its calculation formula is [30]:

$$\tau_{ex} = \frac{d\theta_p^2}{d\omega} = -\frac{\lambda^2}{2\pi c} \frac{d\theta_p}{d\lambda} \quad (9)$$

where, $\theta_p = \text{phase}(\rho)$, ρ is the reflection coefficient or transmission coefficient, and c is the speed of light. The time delay τ_{ex} is calculated to be 2.537 ps. Substituting the obtained value and the values in Table 1 into Formula (8), the linewidth is calculated to be 62.28 kHz. Furthermore, from Equation (8), we can see that the coupling coefficient and grating reflectivity are inversely proportional to the laser linewidth.

Therefore, we developed a three-dimensional finite-element thermal model. This model was used to analyze the thermal field distribution during the crystallization and amorphization of Sb_2S_3 and its consequent impact on the SOA region, specifically at the peak temperatures of each transition. The analysis focused on varying the spacing between the SOA and the external cavity to determine the critical distance required to prevent thermal crosstalk. Table 2 lists the material and thermal parameters employed in the simulation.

Table 2. Main material parameters used in the thermal simulation

Material	k (W/m K)	C_p (J/kg K)	ρ (kg/m ³)
a- Sb_2S_3 [31,32]	0.21	353	4150
c- Sb_2S_3 [31]	0.74	368	4640
Si_3N_4 [33,34]	18.4	774	2750
SiO_2 [35]	1.38	740	2203
ITO [36]	11	1290	7100
Pd [37]	71.8	244	12023

We first simulated the temperature evolution of Sb_2S_3 in response to amorphization and crystallization pulses. Figure 5(a) shows the temperature profile of the Sb_2S_3 during the amorphization process under a 8 V, 177 ns pulse. At the end of the pulse, the peak temperature reaches 1003 K, exceeding its melting point ($T_m \approx 801$ K) and thus enabling complete amorphization upon rapid quenching. For the crystallization process, a lower-amplitude, longer-duration electrical pulse (3 V, 1500 ns) is employed to induce a controlled phase transition. As shown in Fig. 5(b), this pulse raises the Sb_2S_3 temperature to 637 K, which is above its crystallization temperature yet well below its melting point, ensuring a reliable transition to the crystalline state. To further analyze the thermal distribution within the waveguide structure, The temperature profiles across the WBG's yz-plane region are shown in Figs. 5(c) and (d). at 177 ns during amorphization and at 1500 ns during crystallization, respectively. Collectively, the simulation results in Fig. 4 demonstrate that a full phase-change cycle can be completed in less than 7 μs .

Based on the condition that the phase-change material reaches its peak transient temperature of 1003 K, we further theoretically analyzed the thermal crosstalk between the SOA and the external cavity, focusing on how the spacing between them influences the crosstalk intensity. We simulated and calculated the effect of the gap between the external cavity laser (ECL) and the semiconductor optical amplifier (SOA) on the maximum temperature at the cavity-proximal side of the SOA. Figures 6(a) to (e) present the temperature distribution at the cavity-proximal side of the SOA for gaps of 0 nm, 40 nm, 60 nm, 80 nm, and 100 nm, respectively; Fig. 6(f)

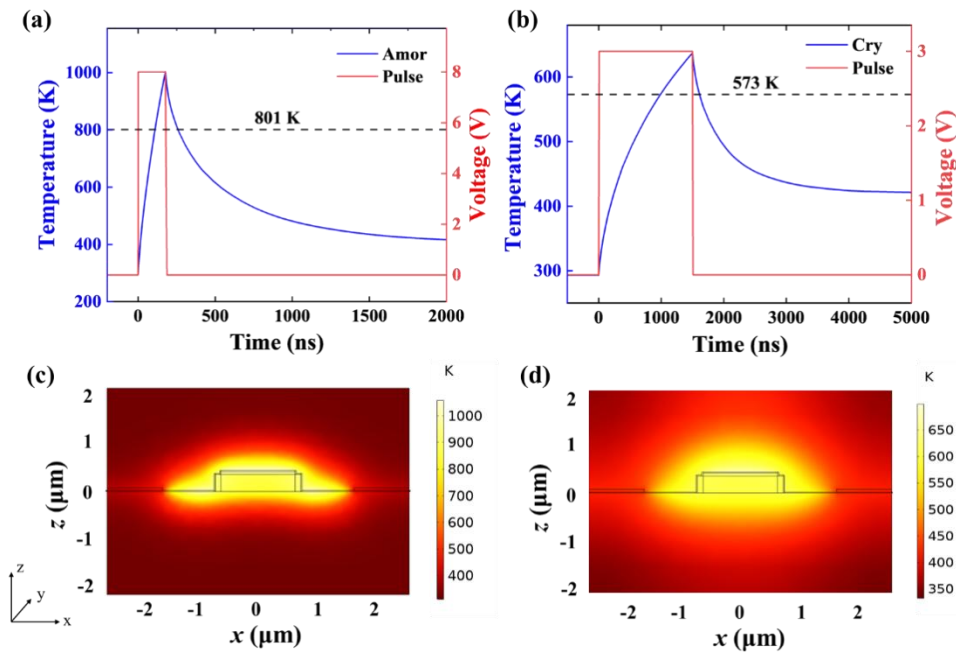


Fig. 5. Temperature evolution of Sb_2S_3 applying (a) a 8-V amorphization pulse when the material is initially crystalline, and (b) a 3-V crystallization pulse when the material is initially amorphous. (c) and (d) show the corresponding cross-sectional temperature distributions of the WBG in the yz-plane under Joule heating.

further summarizes the relationship curve between the gap and the maximum temperature at the cavity-proximal side of the SOA. The results show that when the gap > 0 nm, the temperature rise caused by thermal crosstalk decreases rapidly; when the gap increases to 100 nm, the maximum temperature on the cavity-proximal side of the SOA is approximately 299 K, already approaching room temperature. Furthermore, calculations indicate that at a gap of 20 μm , the maximum temperature at the cavity-proximal side of the SOA drops to room temperature (approximately 293 K), and heating the PCM does not cause thermal crosstalk to the SOA. This demonstrates that under non-contact integration conditions, the thermal crosstalk between the ECL and SOA can be effectively suppressed. However, the physical separation between the SOA and the external cavity may reduce the optical coupling efficiency between the two facets. In practice, to enhance the coupling efficiency, a microlens system can be inserted between the SOA and the external cavity. Alternatively, advanced integration techniques such as photonic wire bonding (PWB) [38] can be employed for efficient facet-to-facet optical coupling. These approaches effectively optimize the optical transmission efficiency of the proposed laser, thereby maintaining high overall performance while suppressing thermal crosstalk.

To enable wavelength tuning across the C-band with a channel spacing of 100 GHz (~ 0.8 nm), a total of 44 tunable states is required. Wavelength switching can be achieved by applying precisely controlled electrical pulse sequences to the phase-change material. These pulses progressively modulate the crystallization fraction of the material from the amorphous to the crystalline state, thereby varying its effective refractive index. During the amorphization process, the relative permittivity of Sb_2S_3 at intermediate states can be calculated by the following formula [39]:

$$\varepsilon_{\text{eff}_a}(\lambda) = \frac{1}{4} \left(\sqrt{8\varepsilon_s\varepsilon_l + ((3f_s - 2)\varepsilon_l + \varepsilon_s(1 - 3f_s))^2} + (2 - 3f_s)\varepsilon_l + \varepsilon_s(3f_s - 1) \right) \quad (10)$$

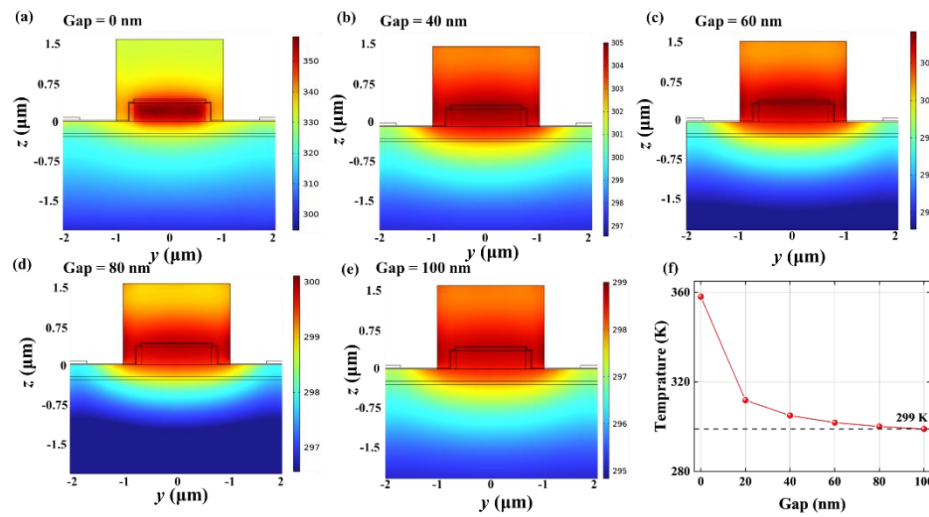


Fig. 6. Temperature distribution at the cavity-proximal side of the SOA under different gap conditions: (a) 0 nm, (b) 40 nm, (c) 60 nm, (d) 80 nm and (e) 100 nm. (f) Relationship curve between the gap and the maximum temperature at the cavity-proximal side of the SOA.

where $\epsilon_{\text{eff}_a}(\lambda)$ represents the equivalent relative permittivity of Sb_2S_3 during the amorphization process, ϵ_s denotes the dielectric constant of the solid state, ϵ_l is the dielectric constant of the liquid state, and f_s indicates the local amorphization fraction. Using this model, the amorphization degree of Sb_2S_3 and its corresponding equivalent refractive index can be quantitatively determined throughout the phase transition.

To achieve reconfigurable tuning of multi-wavelength optical signals, this work discretizes the amorphous-to-crystalline phase transition of Sb_2S_3 into a series of intermediate states with varying degrees of crystallinity. Each intermediate state corresponds to specific dielectric constants and optical refractive indices, enabling precise control of the effective refractive index of optical waveguides. Figure 7(a) presents the calculated relationship between the crystallization/amorphization fraction of Sb_2S_3 thin films and the effective refractive index based on Eq. (9), revealing a linear relationship. The variation in Sb_2S_3 crystallinity is realized by adjusting the duration of externally applied electrical pulses. Figure 7(b) presents the crystallinity results obtained under different voltage pulse conditions, showing the temperature distribution in the crystallized regions of the Sb_2S_3 material (i.e., areas exceeding 573 K). Based on this mechanism, the proposed external cavity laser demonstrates feasibility for continuous and repeatable wavelength tuning in the C-band.

Figure 8 illustrates the dynamic wavelength tuning performance of the proposed tunable laser, realized by precisely controlling the amorphous and crystalline fractions of the phase-change material (PCM) in the external cavity. Simulation results show that the laser output wavelength can be continuously tuned across the C-band (1530 nm–1564.4 nm) with a uniform channel spacing of approximately 0.8 nm between adjacent wavelengths. These results validate the “discretized intermediate-state” phase-transition control strategy, demonstrating its capability to achieve broad, precise, and repeatable wavelength tuning. It’s an effective approach for high-performance tunable lasers in photonic integrated circuits.

We compared the performance of the proposed ECL with other tunable semiconductor lasers used for WDM systems, and the results are shown in Table 3. Compared with the reference schemes, the proposed ECL exhibits advantage in tuning range. Furthermore, the reflective device proposed in this work features low holding power after wavelength tuning. These comprehensive

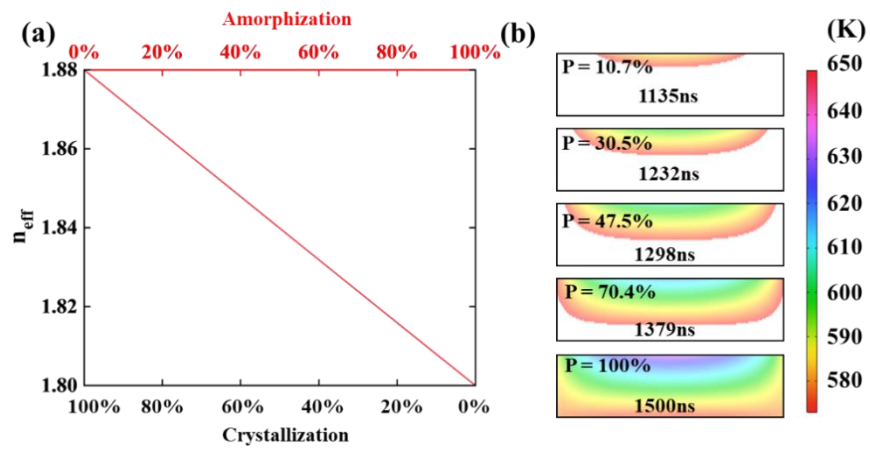


Fig. 7. (a) Relationship between the crystallization/amorphization fraction of the Sb_2S_3 and its effective refractive index. (b) Crystallization of Sb_2S_3 versus voltage pulse duration.

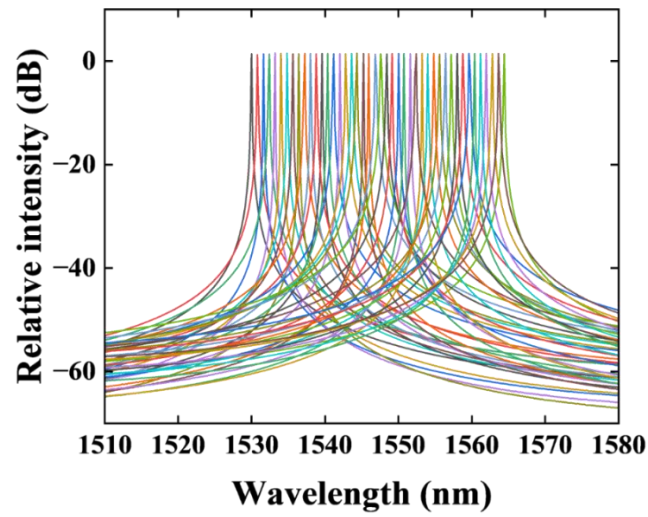


Fig. 8. Laser wavelength tuning via controlled crystallization and amorphization of Sb_2S_3 , with a channel spacing of 0.8 nm.

performance characteristics indicate that this method holds great promise for applications in the field of integrated photonics.

4. Conclusion

In this paper, we propose and demonstrate a novel external-cavity laser based on hybrid integration of a silicon nitride waveguide sidewall Bragg grating and Sb₂S₃ phase-change material. The design fully exploits the high contrast in refractive-index between two states of Sb₂S₃, together with its non-volatile phase-transition characteristics, enabling fast, stable, and reconfigurable tuning of the external-cavity resonance wavelength over a broad spectral range. Simulation results confirm that the laser can cover the entire C-band with a uniform channel spacing of ~0.8 nm. This device offers a technically promising and potentially impactful solution to key challenges in optical communication systems, including precise wavelength control, dynamic resource allocation, and enhanced network flexibility.

Table 3. Performance comparison between the proposed ECL and Other tunable semiconductor lasers for WDM

Reference	Structures	Channel spacing (GHz)	Tuning range (nm)	Linewidth (kHz)	Date Type
[40]	Surface high-order Bragg grating	50	24.4	-	experiment
[41]	High order colliding pulse mode-lock	100	11.5	440	experiment
[42]	V-cavity tunable semiconductor	100	25	-	experiment
This work	Silicon nitride grating hybrid integrated	100	34.4	62.28	simulation

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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